

Identifying High-Value Locations and Best-Selling Products

An Analysis of Coffee Shop Sales Data

Caryn Kesuma Shruti Pillai Nattamon Wattanapenpaiboon
Thunchanok Udomittipong

Table of contents

1 Executive Summary	1
2 Introduction	1
3 Dataset Introduction	1

1 Executive Summary

2 Introduction

The central problem addressed in this report is how a new café can choose a location and initial product offering that maximises early sales and customer engagement. The report focuses on two key objectives: identifying the location with the strongest potential for expansion, and determining which product should be prioritised at launch. Together, these findings will support strategic decision-making for launching a new café with the highest likelihood of early success.

3 Dataset Introduction

To tackle this problem, the analysis uses the Coffee Shop Sales dataset from Maven Analytics, which provides detailed transaction-level information across multiple existing stores. The

dataset includes 149116 rows and 11 columns, representing a large volume of customer purchase activity. Key variables include numeric identifiers such as `transaction_id`, `store_id`, and `product_id`, which uniquely identify each transaction, store, and product. Temporal variables include `transaction_date` (stored as character) and `transaction_time` (stored as a time object), capturing when each purchase occurred. Measures of customer behaviour are represented through `transaction_qty`, a numeric variable indicating the number of items purchased per transaction. Categorical variables (stored as character) such as `store_location`, `product_category`, `product_type`, and `product_detail` describe where the transaction took place and the type of product purchased. Together, these variables provide a rich foundation for analysing sales patterns across locations and product categories.